



Estimating Eligibility for Lung Cancer Screening in Ireland: A Dynamic Modelling Study of Smoking Trends

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Abstract

Background Lung cancer screening (LCS) with low-dose computed tomography (LDCT) reduces mortality. Ireland will likely adopt a national screening programme in the coming years, and evidence on the size of the eligible population is required to inform planning.

Objective The objective was to estimate the number and characteristics of individuals in Ireland eligible for LCS under different international criteria and to project future uptake.

Methods We combined population data from the 2022 Census and subsequent projections with smoking history from the 2017 Eurobarometer survey. Eligibility was assessed under four criteria: Irish pilot (55–74 years, ≥ 20 pack-years, quit < 15 years), US Preventive Services Task Force (USPSTF) 2013, USPSTF 2021, and NELSON. Projections to 2045 accounted for demographic change and smoking trends. Expected uptake was estimated using participation rates from SUMMIT, a large UK LCS implementation trial.

Results In 2022, 245,105 individuals met the Irish pilot criteria, and 345,328 met USPSTF 2021. Eligibility remains broadly stable until the mid-2030s, then declines across all criteria. Relative differences between criteria persist over time, with USPSTF 2021 consistently identifying the largest group. Applying SUMMIT uptake rates, around 36,000 LDCT scans annually would be required at steady state.

Conclusions Approximately a quarter of a million people in Ireland would currently be eligible for LCS, rising to more than 340,000 under broader criteria. Numbers are expected to remain high for the next decade before gradually declining. These findings provide key evidence for service capacity planning and highlight the importance of flexible programme design to meet future demand.

Key Points for Decision Makers

This study estimates that approximately 245,000 people in Ireland were eligible for lung cancer screening in 2022 under national pilot criteria, compared with 345,000 under USPSTF 2021 criteria.

Eligibility remains stable until the mid-2030s and then declines, informing phased rollout and capacity planning.

An estimated 36,000 LDCT scans per year would be expected at steady state using SUMMIT trial uptake rates.

1 Introduction

Lung cancer (LC) remains the leading cause of cancer-related death in Ireland and worldwide. Although it is the second most commonly diagnosed invasive cancer in Ireland, it causes more deaths than any other cancer in both men and women [1]. Survival is poor, primarily owing to late-stage diagnosis: in Ireland, 22% of lung cancer cases are detected at stage III and 39% at stage IV, when curative treatment is rarely feasible [2].

Low-dose computed tomography (LDCT) screening has been shown to reduce lung cancer mortality in high-risk populations. Two large trials—the US National Lung Screening Trial [3] and the European NELSON trial [4]—demonstrated a mortality reduction of 20–24% among people who smoked long-term. On the strength of this evidence, the European Commission has recommended

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LDCT screening as part of a broader cancer prevention strategy across EU Member States [5].

Several countries, including Croatia, Czechia, Poland and Taiwan, have introduced national or regional LDCT screening programmes targeting individuals with a significant smoking history [6–9]. Spain [10] and France [11] have launched their major pilot studies as a precursor to national rollout. In the UK, a phased national rollout of the Targeted Lung Health Check (TLHC) model is underway [12], prioritising high-risk individuals on the basis of smoking history and deprivation level. In Ireland, however, while no formal commitment has been made to full implementation [13], a regional lung cancer screening (LCS) pilot is ongoing and the adoption of a national programme seems likely [14].

1.1 Eligibility Approaches

Often, lung cancer screening programmes define eligibility using rule-based criteria combining age and cumulative smoking exposure (pack-years), with restrictions on time since cessation. These criteria were established in major randomised trials such as NLST and NELSON and subsequently adopted in national guidelines. For example, the 2013 US Preventive Services Task Force (USPSTF) recommendation specified screening for individuals aged 55–80 years with ≥ 30 pack-years who currently smoke or quit within 15 years, while the 2021 update lowered the thresholds to age 50–80 years and ≥ 20 pack-years. The NELSON trial similarly applied age and smoking history criteria with cessation limits.

In recent years, individual risk prediction models such as the PLCom2012 have been proposed as an alternative approach to screening eligibility selection. The International Lung Screening Trial (ILST), a large head-to-head comparison of PLCom2012 with USPSTF 2021 criteria, demonstrated improved sensitivity and more efficient identification of high-risk individuals using the risk-based approach [15]. Risk-based selection is currently implemented in the UK Lung Cancer Screening Programme.

Age–pack-year eligibility rules are operationally simple but may have unintended equity implications. Several studies show that fixed pack-year thresholds can yield lower sensitivity for some racial/ethnic minority groups and may underidentify women who develop lung cancer with lower cumulative smoking exposure, even after the USPSTF 2021 expansion [16–19]. Risk-prediction approaches (e.g. PLCom2012) have been proposed in part to improve efficiency and may mitigate several disparities observed under pack-year-based rules [20]. While such models may improve targeting efficiency and address known sex differences in lung cancer risk, they require individual-level risk factor

data that are not available in aggregated census projections. The present analysis therefore focuses on pack-year and age-based criteria currently used or proposed in Ireland and comparable jurisdictions.

1.2 The Challenge of Estimating Screening Demand

For a national screening programme to be viable, policy-makers must understand the size, characteristics and likely behaviour of the eligible population. This includes forecasting how many people would qualify for LDCT screening under different clinical criteria and how that number might change over time owing to shifts in smoking behaviour and population demographics.

To date, most research on LCS eligibility has relied on static, cross-sectional assessments derived from administrative or survey data, providing little insight into how eligibility may change over time [21, 22]. Studies in Denmark [23], the USA [24], Spain [25] and France [26] have applied various criteria, typically age range, pack-year thresholds and risk criteria, to estimate the number of individuals eligible for screening. These studies offer useful snapshots but lack a dynamic component that accounts for temporal trends in smoking initiation, cessation and intensity. A recent study from Australia [27] estimated LCS eligibility from 2025 to 2030 using individual-level survey data and prediction techniques. This method enables granular projections and is well-suited for estimating the real-world reach of screening programmes. However, it requires a sufficiently large individual-level dataset to ensure precise and reliable predictions, and it is less flexible for simulating alternative eligibility criteria or exploring future policy scenarios.

In the Irish context, no published study has attempted to estimate the size of the LCS-eligible population. Smoking remains the predominant risk factor for LC in Ireland, accounting for an estimated 76% of incident cases [28]. Although national smoking prevalence has declined steadily, from 32% in 2015 to 18% in 2023 [29], eligibility for LDCT is shaped by historical smoking exposure. Many individuals in the screening-eligible age group (55–74 years) began smoking in an era of much higher prevalence and intensity. Thus, falling smoking rates do not immediately translate into lower screening demand.

1.3 Aims and Objectives

This study aims to provide the first dynamic, Ireland-specific estimates of eligibility for LDCT lung cancer screening. Using data from the 2022 Census [30], the 2017 Eurobarometer survey [31] and Central Statistics Office (CSO) population projections [32], we constructed a discrete-time Markov model that simulates transitions in smoking behaviour and tobacco exposure over time. Eligibility was assessed under

four international criteria currently used in research and policy.

Our objectives are: first, to estimate the number of individuals eligible for LDCT screening in 2022, stratified by age, sex and region; second, to project eligibility trends to 2045 under five plausible smoking scenarios; and third, to estimate potential screening uptake on the basis of participation patterns observed in the UK SUMMIT study. These projections are intended to inform national screening policy and resource planning, ensuring alignment between eligibility thresholds, service capacity, and evolving public health risk.

2 Methods

2.1 Study Design and Overview

We developed a discrete-time Markov cohort model to estimate the annual number of individuals in Ireland who would be eligible for LDCT screening between 2022 and 2045. The model integrates demographic projections, smoking prevalence, and cumulative tobacco exposure. Eligibility projections are constructed as a repeated cross-sectional population accounting framework: annual CSO population counts by age and sex already incorporate mortality at the aggregate group level. Within each projected annual population snapshot, smoking status from Markov model and cumulative exposure distributions were applied to estimate screening eligibility.

Eligibility was assessed under four established criteria: Irish pilot criteria (age 55–74 years, ≥ 20 pack-years, quit ≤ 15 years), USPSTF 2013 (age 55–79 years, ≥ 30 pack-years, quit ≤ 15 years); USPSTF 2021 (age 50–79 years, ≥ 20 pack-years, quit ≤ 15 years); and NELSON trial (age 50–74 years, ≥ 15 pack-years, quit ≤ 10 years). The Irish pilot criteria reflect those used in the ongoing Lung Health Check pilot [14]. Eligibility in all definitions required current or former smoking status (within the specified quit window) and meeting the corresponding pack-year threshold.

2.2 Data Sources

Three main datasets informed the model. The 2022 Census provided population counts by age, sex and NUTS-3 region, including self-reported smoking status categorised as people who currently smoke, people who formerly smoked or people who have never smoked [30]. Population projections from the CSO covered the period 2022–2045 and included age- and sex-specific estimates under the medium-migration scenario ('M2') [32]. Individual-level smoking histories were derived from Eurobarometer 2017 ($n = 1008$ for Ireland), which provided data on age at initiation and cessation,

intensity of use and duration of smoking. Although the Eurobarometer data were collected in 2017, they were used to estimate the distributional shape of cumulative smoking exposure within age–sex strata. Because pack-years represent long-term accumulated exposure, we assumed that the within-stratum distribution evolves gradually over short intervals (2017–2022). To improve the precision of age- and sex-specific transition probabilities, the broader EU-28 sample was used for model estimation [31]. For detailed information, see Online Resource 1.

2.3 Estimating Pack-Year Histories

For each respondent in the Eurobarometer dataset, pack-years were calculated as:

$$\text{Pack - years} = \text{ECPD} \times \text{years smoked}/20,$$

where effective cigarettes per day (ECPD) was adjusted using an exponential uptake model to reflect progressive increases in smoking intensity from initiation (see Online Resource 2).

Pack-year distributions were summarised by age and sex and assumed to follow a log-normal distribution, which provided a good empirical fit. These were mapped onto Census 2022 smoking prevalence data to estimate the number of individuals within each demographic group exceeding 15, 20 or 30 pack-years.

2.4 Baseline Eligibility Estimation

Baseline eligibility for 2022 was estimated by applying the age–sex-specific, log-normal pack-year distributions of people who currently smoke and people who formerly smoked. Evidence suggests that log-transformed measures of smoking exposure are stronger predictors of lung cancer risk than untransformed pack-years [33]. The proportion exceeding each pack-year threshold was calculated using the survival function of the log-normal distribution:

$$\text{Proportion eligible} = 1 - \Phi\left(\frac{\ln(X) - \mu}{\sigma}\right),$$

where X is the eligibility threshold defined in pack-years, μ is the mean of log pack-years estimated for each age–sex cohort; and σ is the standard deviation of the log pack-years, reflecting variability in smoking exposure within the cohort.

2.5 Modelling Smoking Transitions

Census 2022 data provided the baseline shares of people who currently smoke, people who formerly smoked and

people who have never smoked by age and sex. Using a Markov model, these shares were projected annually from 2022 to 2045. In the model, individuals transitioned between three mutually exclusive states—currently smoking, formerly smoking and never having smoked—using the following matrix for each age-sex cohort (1):

$$M = \begin{bmatrix} 1 - \text{quit_rate} & \text{quit_rate} & 0 \\ \text{initiation_rate} & 1 - \text{quit_rate} & 0 \\ \text{initiation_rate} & 0 & 1 - \text{initiation_rate} \end{bmatrix} \quad (1)$$

At each cycle, the current distribution of individuals across states is multiplied by this matrix to calculate the distribution for the next time step. For the youngest cohort (aged 15–19 years), an annual inflow rate of 0.02 into the “never having smoked” state was applied to reflect reduced smoking initiation rates starting from the baseline. Similar adjustments for the subsequent age groups were made, accounting for the change in smoking take-up and the increasing share of people who have never smoked over the model timeline (see details in Online Resource 3). The shares of people who currently smoke and those who formerly smoked in these cohorts were proportionally reduced, and the shares were renormalised.

Age-dependent transition probabilities for initiation and cessation were estimated via logistic regression using EU-28 Eurobarometer data. Self-reported information on age, smoking status, age at initiation and age at cessation allowed us to identify individuals who had stopped smoking within the past year and those who had started smoking within the past year. Two separate models were fitted: one predicting the probability of quitting among people who currently smoke and one predicting the probability of initiation among people who have never smoked. Both models included age (continuous) and gender (binary) as predictors. Predicted probabilities were derived across the age span from 15 to 80 years using the fitted coefficients.

2.6 Projecting Smoking Exposure and Eligibility

To model eligibility in future years, age–sex-specific pack-year distributions were applied to dynamically updated smoking status proportions. We introduced an attenuation factor to the log-mean of the pack-year distribution to account for reductions in smoking intensity and earlier cessation, consistent with Irish data. The attenuation rate was applied uniformly across age and sex owing to limited evidence to support sex-specific projections of future exposure decline. The log-variance was held constant across all years and scenarios. The attenuated pack-years for a given cohort were calculated as:

$$\text{Attenuated pack - years} = \text{initial pack - years} \times (1 - r)^n,$$

where r is the annual attenuation rate and n the number of years since the baseline year (2022).

2.7 Scenario Analysis

To address uncertainty in future smoking behaviours and its effect on eligibility, we modelled five scenarios. These varied in two dimensions: whether smoking initiation continued beyond 2022 and the extent of attenuation in smoking exposure over time. The base-case assumed continued initiation and a modest (1%) annual decline in exposure. Alternative scenarios included higher attenuation (3%), no attenuation, and zero smoking initiation from 2022 onwards. These variations allowed us to explore the implications of different public health trajectories on future screening demand.

2.8 Model Assumptions

Several simplifying assumptions were required owing to data and model constraints. First, transition probabilities for smoking behaviour were based on EU-28 data, as the Irish subsample alone was insufficiently powered. Second, attenuation of exposure was applied uniformly across age and sex, without adjusting for regional or socioeconomic differences in smoking behaviour.

Third, smoking-related mortality was not modelled as a separate process by smoking status. Instead, annual population totals were taken from CSO demographic projections, which incorporate mortality at the population level. Because these projections do not differentiate survival by smoking status, the model does not explicitly simulate higher mortality among people who currently smoke or those with higher smoking intensity within cohorts. Introducing smoking-specific mortality in addition to CSO projections would require modelling deaths by smoking status and reconciling these with official population totals, which could otherwise lead to double-counting mortality effects. Consequently, the model reflects aggregate mortality trends but may modestly overestimate the number of individuals with long-term, high-intensity smoking exposure surviving to older ages. However, given that national population projections already embed the historical mortality impact of smoking, this limitation is unlikely to substantially alter baseline eligibility estimates.

Fourth, transition probabilities were assumed constant within age–sex cohorts over time. Finally, the model does not retain individual smoking histories, meaning that repeated initiation or cessation events are possible within the framework. However, given the long-term nature of pack-year accumulation, this is unlikely to substantially affect eligibility projections over the time horizon modelled.

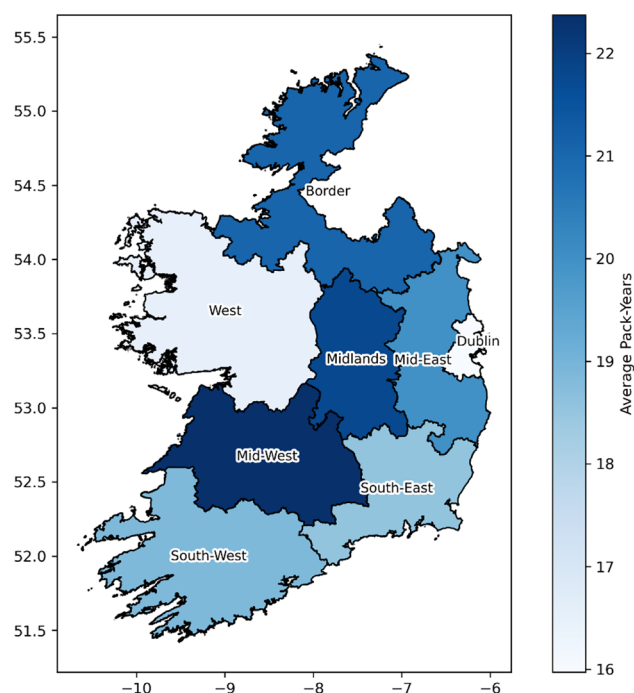


Fig. 1 Regional variation in average pack-years across Ireland (Eurobarometer 2017 data). The figure shows regional variation in average smoking exposure in pack-years among people who have ever smoked in Ireland. Darker shades indicate higher mean pack-year histories, with values ranging from 16 to more than 22 pack-years across regions. The figure illustrates the role of accounting for smoking exposure when planning for a national lung cancer screening programme

These limitations were explored in scenario and sensitivity analyses. Importantly, they are not expected to materially influence the baseline estimates for 2022, which form the primary basis for immediate policy planning.

2.9 Outcomes

The primary outcome was the estimated number of individuals eligible for LDCT screening in 2022, stratified by age, sex, and region, and evaluated under each screening criterion. Secondary outcomes included projected eligibility through 2045 across all scenarios, as well as estimated demand for LDCT scans based on uptake rates observed in the UK SUMMIT study [34, 35].

3 Results

3.1 Smoking Status and Cumulative Smoking Exposure in Ireland

In 2022, 16% of adults in Ireland reported current smoking, and a further 24% reported having stopped smoking.

Regional estimates of average cumulative smoking exposure, derived from individual-level data from the Irish subsample of the 2017 Eurobarometer survey, indicate substantial geographic variation in pack-years. These regional differences highlight that planning for lung cancer screening should incorporate cumulative smoking exposure, not solely population density or smoking prevalence. The highest mean exposures were observed in the Border, Midlands and Mid-West regions (both > 20 pack-years), while lower levels were recorded in Dublin and West (Fig. 1).

Figure 2 displays complementary cumulative distribution functions (CCDFs) of pack-year exposure for people who currently smoke and those who have stopped smoking, stratified by age and screening eligibility thresholds. Among people who currently smoke across all ages, more than 60% exceeded 15 pack-years, and approximately 45% exceeded 20 pack-years, suggesting a substantial proportion may meet LDCT screening thresholds. Among people who have stopped smoking, the distribution declined more sharply, indicating lower cumulative exposure and earlier cessation.

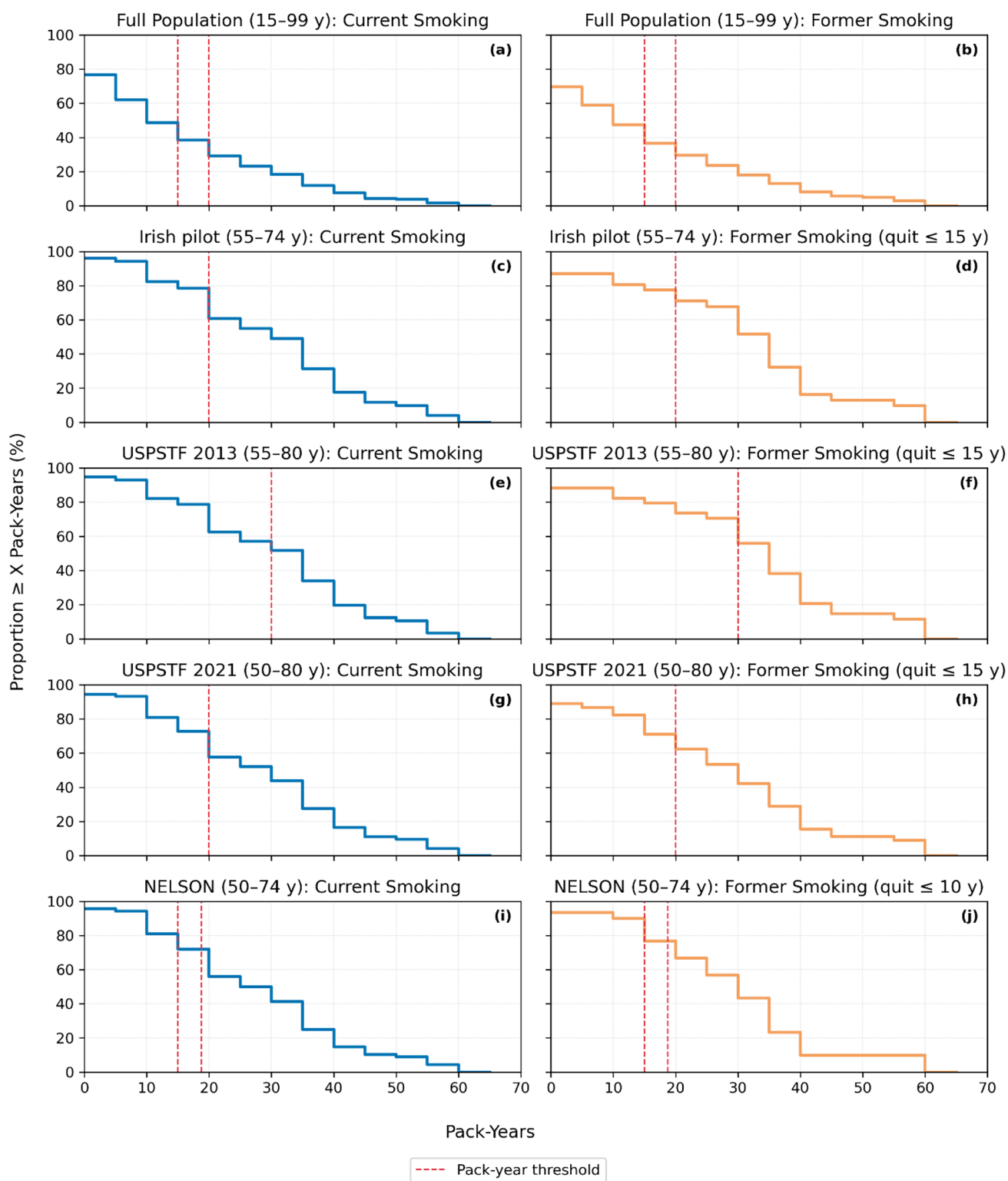
When restricted to individuals aged 55–74 years, the proportion meeting pack-year thresholds remained high for people who currently smoke and substantial for those who stopped smoking under both Irish and NELSON criteria. These findings confirm that a sizeable proportion of older adults in Ireland have a history of smoking consistent with international screening guidelines.

3.2 Baseline Eligibility for Lung Cancer Screening

Table 1 presents modelled estimates of screening eligibility in 2022 under four criteria. Nationally, 245,105 individuals met the Irish pilot criteria (≥ 20 pack-years, quit within 15 years). Broader definitions yielded higher estimates: 262,238 under USPSTF 2013, 345,328 under USPSTF 2021 and 293,161 under the NELSON criteria.

Eligibility was consistently higher among men (54–56%) across all definitions, reflecting historical differences in smoking duration and intensity. Most eligibility was concentrated in those aged 55–74 years, particularly between 55 and 64 years. Age distributions within sex were broadly similar; however, female individuals were modestly more represented in the oldest eligible age bands (e.g. 75–79 years under USPSTF 2013), consistent with longer life expectancy. Lowering the age threshold to 50 years (USPSTF 2021) increased representation in the 50–54-year age group for both sexes.

Regionally, Dublin accounted for the highest absolute number of eligible individuals ($n = 63,382$), followed by the South-West and Mid-East. However, regional variation in eligibility rates as a share of the population was modest (± 2 percentage points), suggesting broad geographic reach



◀**Fig. 2** CCDF of pack-years by screening eligibility criteria (Irish subsample of Eurobarometer 2017). Complementary cumulative distribution functions (CCDFs) of pack-year exposure among people who currently smoke (left column) and people who stopped smoking within the quit-time eligibility window (right column), stratified by screening criteria. Panels show (a, b) full population, (c, d) Irish pilot criteria (55–74 years, ≥ 20 pack-years, quit < 15 years), (e, f) USPSTF 2013 (55–80 years, ≥ 30 pack-years, quit < 15 years), (g, h) USPSTF 2021 (50–80 years, ≥ 20 pack-years, quit < 15 years), and (i, j) NELSON (50–74 years, > 15 cigarettes/day for 25+ years (~ 18.75 pack-years) or > 10 /day for 30+ years (~ 15 pack-years), assuming 20 cigarettes per pack). Vertical dashed red lines indicate the pack-year thresholds used in each criterion. USPSTF, US Preventive Services Task Force; NELSON, Dutch–Belgian Randomized Lung Cancer Screening Trial.

would be required for equitable programme implementation. The male-to-female eligibility ratio was lowest in Dublin, consistent with more balanced smoking histories in urban populations.

3.3 Transition Probabilities: Smoking Initiation and Cessation

Figure 3 reports the probability of smoking take-up or cessation estimated through the logistic regression using EU-wide Eurobarometer data. It is not surprising that smoking cessation rates sharply decline with age, peaking during adolescence and early adulthood. In contrast, initiation probabilities fall modestly with age, suggesting that older individuals are less likely to take up smoking. Gender differences are minimal in the quit rate and are higher in the initiation rate among young men compared with women.

3.4 Projected Smoking Trends to 2045

Projected smoking behaviour under the base-case scenario is shown in Fig. 4. Current smoking prevalence is expected to decline steadily from 16% (0.85 million people) in 2022 to 6% (0.43 million) in 2045. The share of if individuals who used to smoke is projected to remain stable at approximately 25% (≈ 1.4 million), while the proportion of people who never smoked increases from 53% (2.75 million) to over 66% (4 million people) by 2045.

These trends reflect ongoing reductions in smoking initiation among younger adults and continued cessation among older groups. The alternative ‘no initiation’ scenario, where smoking uptake ceases entirely from 2022, suggests similar but more pronounced trends: prevalence increases more rapidly for people who never smoked. While these prevalences trend towards the Tobacco-Free Ireland goal of reducing smoking prevalence to less than 5% [36], the current trajectory suggests the proportion of people who smoke will remain above 10% until 2035. The persistence of this group has implications for LCS. Over the next 20 years,

younger cohorts currently experiencing significant declines in smoking prevalence will age into the 45–55-year groups but remain below the eligibility threshold for screening. By 2045, the remaining 7% of those who currently smoke will primarily be individuals who were aged 25–44 years in 2025. Smoking in the 25 to 34-year age group has been relatively high, however, dropping from 32% in 2015 to 20% in 2024 [37].

3.5 Projected Eligibility for LDCT Screening (2022–2045)

Figure 5 presents projections of LDCT eligibility under four screening criteria (Irish pilot, USPSTF 2013, USPSTF 2021 and NELSON) and five future smoking scenarios. Eligibility projections are highly sensitive to the attenuation assumptions and serve as hypothetical scenarios to explore potential impacts of LCS. For illustration, the 1% attenuation rate is approximately equivalent to smoking 0.2 fewer cigarettes per day over 20 years, or about 2–3 months less of smoking at one pack per day, given a 20 pack-year history. The 3% attenuation rate is approximately equivalent to smoking around 0.6 fewer cigarettes per day over 20 years, or about 7–8 months less of smoking at one pack per day.

In the ‘no attenuation’ scenario, where smoking intensity remains constant over time, eligibility increases steadily across all strategies. For the Irish pilot criteria, the number of eligible individuals rises by nearly 20% by the mid-2030s, reflecting population growth predicted by the CSO (see Online Resource 4). However, this trajectory assumes that smoking intensity remains fixed at current levels, which is increasingly unlikely given ongoing cessation efforts and behavioural shifts towards reduced tobacco use.

More plausible scenarios involve gradual declines in smoking intensity. The base-case scenario (1% annual attenuation in pack-year exposure) results in a plateau around 2035, followed by a gradual decline in eligibility. The impact of preventing smoking initiation is limited in the short term owing to the delayed nature of eligibility accumulation, though effects become more visible after 2035.

Under higher attenuation (3% annually), eligibility declines more sharply, with reductions of up to 36% projected by 2045 for the Irish pilot criteria.

All above suggests that suppressing smoking initiation alone, while impactful over the long term, has only modest effects within the current projection window, as most individuals who have never smoked will not reach screening age before 2045.

In sensitivity analyses, cumulative pack-years were inflated by 5%, 10% and 20% to account for the potential underreporting of smoking exposure. This was implemented as a proportional shift in the log-mean of the pack-year distribution. Under these assumptions, projected eligibility

Table 1 Estimated eligibility for LDCT screening in Ireland, 2022 (by region, age and criterion)

Strategy (PY, quit < x years)	Region	Total eligible	Male (%)	Female (%)	50–54 (%)	55–59 (%)	60–64 (%)	65–69 (%)	70–74 (%)	75–79 (%)
<i>Both sexes</i>										
Irish pilot (≥ 20 PY, < 15 years)	State	245,105	54.6	45.4	–	34.6	27.7	21.1	16.6	–
	<i>Border</i>	21,541	56.2	43.8	–	33.1	27.0	22.2	17.7	–
	<i>Dublin</i>	63,382	52.6	47.4	–	34.8	27.8	21.0	16.5	–
	<i>Mid-East</i>	34,275	55.1	44.9	–	36.2	27.3	20.5	16.1	–
	<i>Mid-West</i>	26,129	54.5	45.5	–	33.9	27.7	21.2	17.2	–
	<i>Midland</i>	15,675	55.8	44.2	–	34.9	28.1	21.1	16.0	–
	<i>South-East</i>	24,273	55.1	44.9	–	35.2	28.0	20.7	16.1	–
	<i>South-West</i>	35,742	55.5	44.5	–	35.1	27.5	21.1	16.3	–
	<i>West</i>	24,088	55.6	44.4	–	32.7	28.0	21.6	17.8	–
		262,238	54.5	45.5	–	31.7	25.4	19.5	15.5	7.8
USPSTF 2013 (≥ 30 PY, < 15 years)										
USPSTF 2021 (≥ 20 PY, < 15 years)		345,328	53.9	46.2	23	24.6	19.6	15.0	11.8	6.0
NELSON (≥ 15 PY, < 10 years)		293,161	54.1	45.9	29.3	24.8	20.2	14.8	10.9	–
<i>Male</i>										
Irish (≥ 20 PY, < 15 years)		133,936	–	–	–	34.8	27.5	21.5	16.2	–
USPSTF 2013 (≥ 30 PY, < 15 years)		142,943	–	–	–	32.1	25.3	19.9	15.3	7
USPSTF 2021 (≥ 20 PY, < 15 years)		185,952	–	–	22.2	25.1	19.8	15.4	11.7	5.8
NELSON (≥ 15 PY, < 10 years)		158,693	–	–	29.7	24.4	20.7	14.7	10.5	–
<i>Female</i>										
Irish (≥ 20 PY, < 15 years)		111,169	–	–	–	34.4	27.9	20.6	17.1	–
USPSTF 2013 (≥ 30 PY, < 15 years)		119,295	–	–	–	31.3	25.5	19.0	15.9	8.3
USPSTF 2021 (≥ 20 PY, < 15 years)		159,377	–	–	24.0	24.0	19.4	14.4	11.9	6.3
NELSON (≥ 15 PY, < 10 years)		134,468	–	–	28.9	25.3	19.6	14.9	11.3	–

Estimated number and distribution of individuals eligible for lung cancer screening in Ireland under alternative eligibility criteria: Irish pilot programme (≥ 20 pack-years, quit < 15 years), USPSTF 2013 (United States Preventive Services Task Force, ≥ 30 pack-years, quit < 15 years), USPSTF 2021 (≥ 20 pack-years, quit < 15 years) and NELSON trial (Dutch–Belgian Randomized Lung Cancer Screening Trial, ≥ 15 pack-years, quit < 10 years). Results are shown overall and by sex and age group, with regional breakdowns for the Irish pilot criteria shown in italics. USPSTF, US Preventive Services Task Force; NELSON, Dutch–Belgian Randomized Lung Cancer Screening Trial.

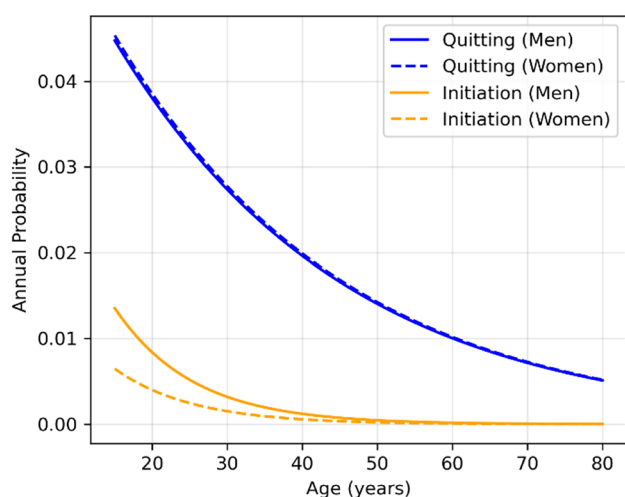


Fig. 3 Quitting and initiation rates by age and gender using Eurobarometer data 2017. Estimated annual probability of smoking initiation and quitting by age and gender in Ireland. Quitting probabilities are shown in blue (solid = men, dashed = women), while initiation probabilities are shown in orange (solid = men, dashed = women). Both initiation and quitting decline steadily with age, with initiation probabilities approaching zero by middle age

under the Irish pilot criteria increased modestly across the time horizon (Online Resource 5), while overall trends remained unchanged. As eligibility assessment in screening programmes also relies on self-reported smoking history, similar reporting biases may arise during real-world recruitment.

3.6 Potential Uptake and Demand for LDCT Scans

Having established that the eligibility may vary over time depending on smoking trends, the demand on screening resources depends on participation. Evidence from the countries with LCS in place suggests that the uptake rates vary between 5 and 15% in the USA [38–40] and 41% in China [41]. It is natural to expect that Ireland will be no exception, with only a share of the eligible population undergoing LDCT.

In the UK, The SUMMIT trial reported a 31.7% uptake of invitations, with 63.0% [34] attending a lung health check, and 70% of attendees proceeded to LDCT scanning [35]. Applying these proportions to the projected 2025 Irish pilot-eligible population ($n = 257,000$), an estimated 81,500

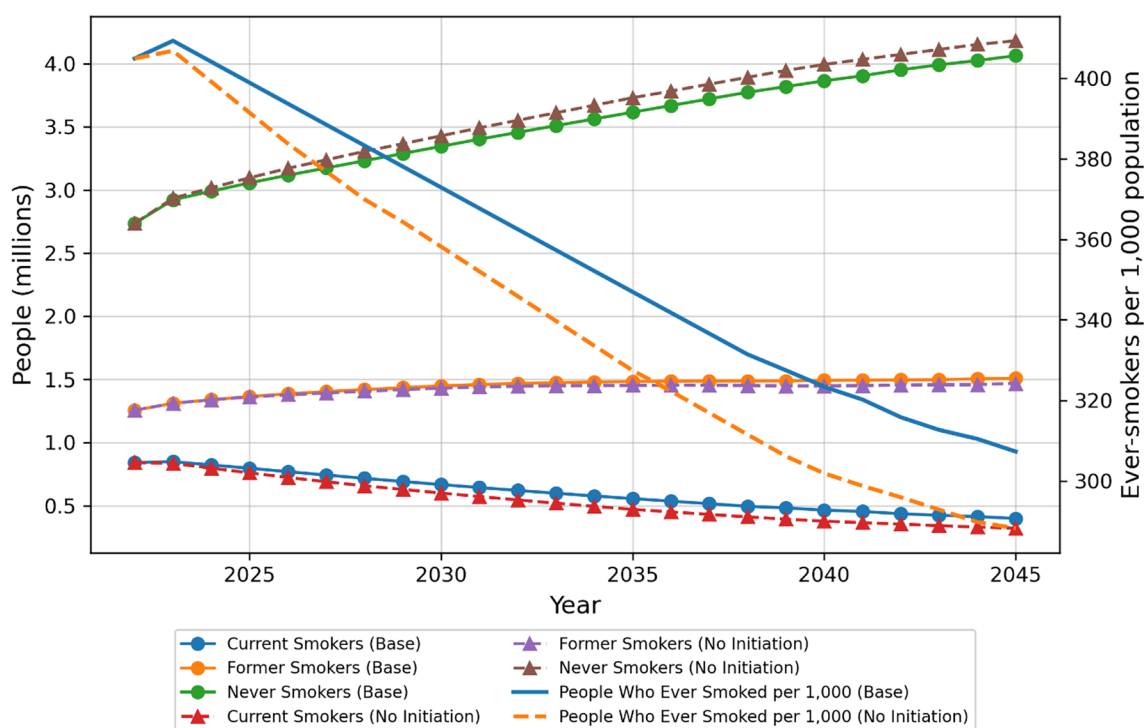


Fig. 4 Projected proportions of people who currently smoke, people who have stopped smoking and people who have never smoked over time. Projected population by smoking status in Ireland, 2022–2045, under two scenarios: base-case (solid lines) and no initiation (dashed lines). The base-case assumes continued smoking initiation, while the no-initiation scenario removes individuals who start smoking from entering the population. Results are shown for people who currently smoke (blue), people who have stopped smoking (orange/purple) and

people who have never smoked (green/brown). The no-initiation scenario leads to a sharper decline in people who smoke and a corresponding increase in those who have never smoked over time, while the number of people who have stopped smoking remains largely stable. Rates among people who have ever smoked (those who currently smoke and those who have stopped smoking per 1000 population) are displayed to provide population-level context and show a gradual decline over time, particularly under the no-initiation scenario.

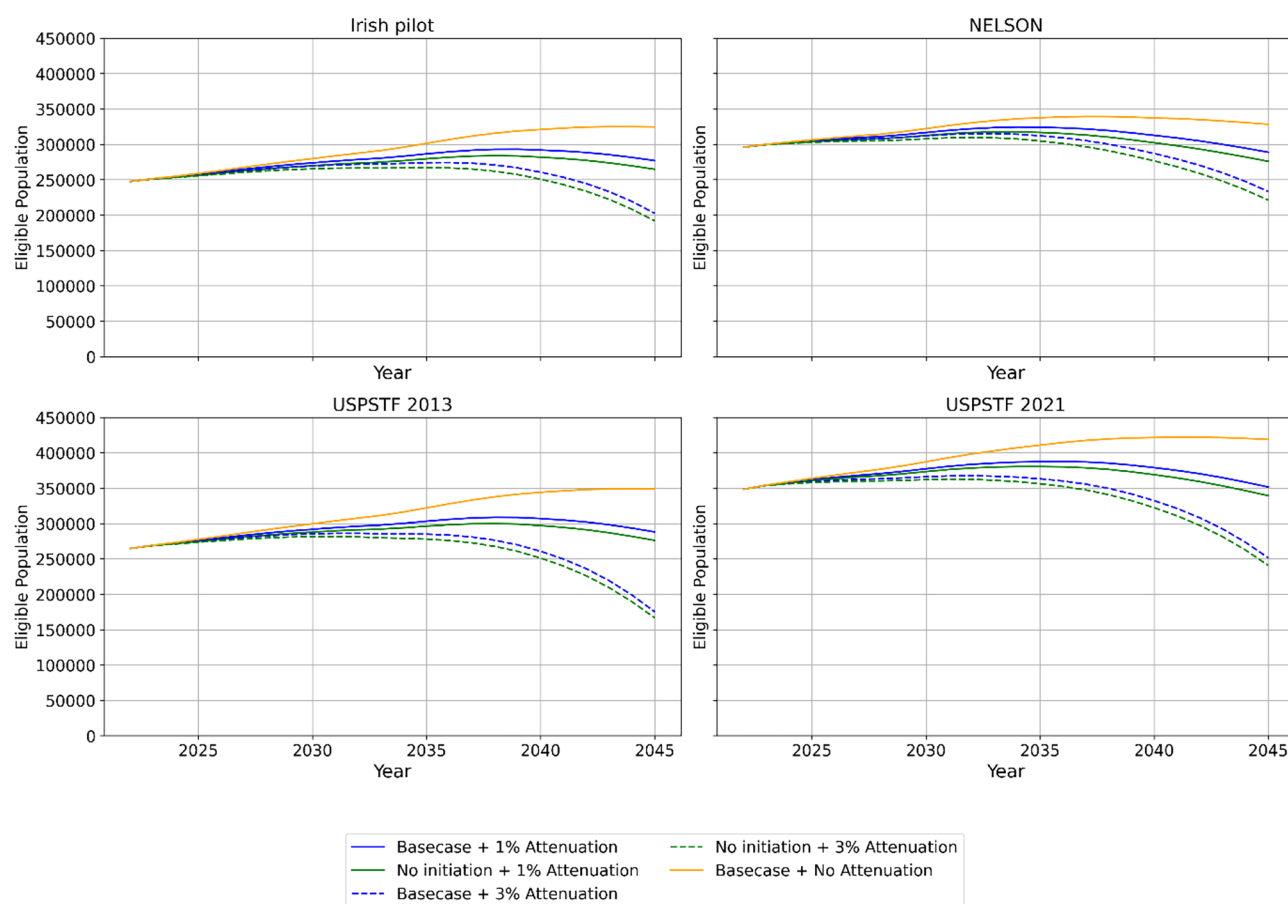


Fig. 5 Projected eligibility for LDCT screening under four criteria and five smoking scenarios, 2022–2045. Projected eligible population for lung cancer screening in Ireland under four criteria: Irish pilot (≥ 20 pack-years, quit < 15 years), NELSON (≥ 15 pack-years, quit < 10 years), USPSTF 2013 (≥ 30 pack-years, quit < 15 years) and USPSTF 2021 (≥ 20 pack-years, quit < 15 years). Projections are shown from 2022 to 2045 under different scenarios: base-case with no atten-

uation (orange), base-case with 1% annual attenuation (blue solid), base-case with 3% annual attenuation (dashed blue), no initiation with 1% attenuation (green solid) and no initiation with 3% attenuation (dashed green). Attenuation reflects a gradual decline in effective smoking exposure over time. USPSTF, US Preventive Services Task Force; NELSON, Dutch–Belgian Randomized Lung Cancer Screening Trial.

individuals would respond to an invitation, 51,300 would attend a lung health check, and approximately 35,900 would undergo LDCT screening, corresponding to an effective uptake rate of 13.6% of those eligible.

Under the broader USPSTF 2021 criteria, projected annual demand for LDCT scans would exceed 50,000. Even assuming lower-end uptake rates (e.g. 5%, as reported in several US cohorts [40]), the number of scans required annually would range from 13,000 to 17,000, depending on the eligibility definition applied.

4 Discussion

This study provides the first Ireland-specific estimates of eligibility for LDCT screening using a dynamic modeling framework. In 2022, an estimated 250,000 adults met the national pilot criteria, which is equivalent to around 7% of the population and nearly one in three individuals aged 55–74 years. Despite a continuing decline in smoking prevalence, the size of the eligible population is projected to remain stable or increase slightly until the mid-2030s, before gradually declining. This reflects the enduring effects of historical smoking and the ageing of previously exposed cohorts [42].

Expanding the eligibility definition to reflect international guidelines, such as the 2021 USPSTF criteria, would

increase the size of the eligible population by approximately 100,000. This expansion would primarily affect women, younger individuals, and those with lower cumulative smoking exposure, consistent with global trends toward more inclusive screening thresholds.

4.1 Implications for Policy and Implementation

Assuming uptake rates similar to the UK SUMMIT study, Ireland would require capacity for around 36,000 LDCT scans annually. While this is a substantial addition to radiology services, it is within the general scale of population screening activity seen in other programmes. For example, BreastCheck has historically delivered between 90,000 and 100,000 mammograms annually [43]. Nonetheless, LDCT requires greater imaging resources, diagnostic triage and radiological oversight, which must be factored into service planning.

Participation is likely to be a key determinant of impact. Although organised screening programmes in Ireland, such as BreastCheck, have historically achieved relatively high uptake, participation has declined in recent years, with coverage falling below 70% even prior to the pandemic [43]. However, lung cancer screening differs in important respects, including the stigma associated with smoking and the requirement for eligibility assessment based on smoking history. BowelScreen may provide a more relevant benchmark: it is open to both sexes yet only achieves ~30% uptake, with pronounced differences by age, sex and socioeconomic position [44]. Barriers to participation include low awareness, stigma, emotional distress and logistical complexity [45, 46]. Individuals at highest risk of lung cancer, including those with substantial cumulative smoking exposure, longer duration of smoking, family history or socioeconomic disadvantage, may be less effectively reached by screening services due to structural and access-related barriers.

International evidence suggests that targeted interventions can improve uptake. Strategies such as GP-endorsed invitations, pre-screening telephone reminders, patient navigation and mobile outreach have shown effectiveness across diverse populations [47–49]. However, gains are often modest and not always equitably distributed. For example, GP endorsement alone has not consistently reduced inequalities in screening participation [50]. Broader, community-informed approaches, including culturally appropriate messaging and co-designed interventions, are likely to be required to address persistent access gaps.

4.2 Addressing Equity and Data Limitations

To ensure equitable implementation, a robust monitoring infrastructure is needed. Current cancer screening programmes in Ireland collect limited sociodemographic data,

hampering the ability to assess and address inequalities. Most do not routinely capture variables such as ethnicity, education or housing status [44]. Integration of unique health identifiers and enhanced data linkage between screening and administrative datasets could enable more granular equity monitoring and targeted outreach.

The projected persistence of eligibility in regions with historically high smoking exposure, such as the Border and Midland areas, reinforces the need for geographically distributed services. Screening rollout should be aligned with regional smoking burden, population need and primary care engagement rather than limited to urban centres.

4.3 Strengths and Limitations

This is the first published study to estimate eligibility for LDCT screening in Ireland, and among the first internationally to project dynamic eligibility over time. A key strength is the integration of multiple data sources—Census 2022, CSO projections and Eurobarometer microdata—to estimate exposure and simulate behaviour. By incorporating transitions in smoking status and modelling attenuation in pack-year accumulation, the approach captures how both individual behaviour and population structure shape future eligibility.

However, several limitations should be noted. First, smoking transition probabilities were estimated using EU-level data owing to the limited sample size in the Irish Eurobarometer subsample. Nonetheless, baseline smoking prevalence in the model closely matched Census 2022 observations, and the Eurobarometer has been previously validated for this purpose [16, 31].

Second, while pack-year thresholds are a practical proxy for risk stratification, self-reported smoking histories are susceptible to recall bias, and many lung cancer cases occur outside defined screening groups. Underreporting of smoking could lead to downward bias in the estimated number of individuals meeting eligibility thresholds. However, eligibility determination in lung cancer screening programmes also relies on self-reported smoking history. As such, the model reflects the same information source likely to be used in programme implementation. Any additional bias would depend on whether the degree of misreporting differs between survey respondents and individuals presenting for screening, which is difficult to quantify. The pack-year distributions were estimated using 2017 survey data and applied to Census 2022. While cumulative exposure patterns are unlikely to change abruptly over short intervals, shifts in smoking intensity or cessation patterns between 2017 and 2022 could modestly affect the estimated exposure distribution.

The attenuation of cumulative exposure was applied uniformly across sexes owing to the limited evidence. Importantly, sex-specific differences in smoking status were

incorporated through separate baseline pack-year distributions and sex-specific initiation and cessation transition probabilities. The uniform attenuation assumption, therefore, affects only the projected rate of decline in smoking intensity (i.e. cumulative exposure) and not the evolution of smoking prevalence by sex. If future reductions in smoking intensity differ substantially between males and females, eligibility projections may be modestly over- or underestimated within specific subgroups.

Third, the use of a Markov model imposes a memoryless structure that does not retain full individual histories, allowing for hypothetical re-initiation after quitting. However, this limitation is unlikely to affect projections meaningfully within the screening age range.

4.4 Future Research

Future work could consider individual-level microsimulation models that incorporate comorbidities, risk scores, and additional exposures to support more personalised screening strategies. Alongside quantitative modelling, qualitative research is needed to understand public perceptions of LDCT screening in Ireland, including barriers related to stigma, acceptability and engagement.

From a systems perspective, there is a clear need for improved data infrastructure. Routine linkage of screening records to sociodemographic and primary care data would enable real-time monitoring of equity and support targeted invitation strategies. Finally, embedding screening within a broader tobacco control framework through coordinated referral to cessation services and public health messaging will be essential to maximise long-term benefits.

5 Conclusions

LDCT screening has the potential to reduce lung cancer mortality by identifying the disease earlier in high-risk populations. This study offers the first dynamic, Ireland-specific projections of screening eligibility and highlights the persistence of screening need over the coming decades. Implementation planning must account for evolving smoking trends, service capacity and public participation. The impact of screening will depend not only on technical feasibility but also on equitable engagement and alignment with broader tobacco control efforts.

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Declarations

Conflict of interest The authors declare no conflicts of interest.

Ethics approval This study used publicly available, de-identified datasets (CSO Census 2022, CSO PEC19 projections, Eurobarometer 2017). Ethical approval was not required.

Consent to participate Not applicable.

Consent for publication Not applicable.

Data availability All data are publicly available: CSO Census 2022 (Table F4042), CSO PEC19 population projections, and Eurobarometer 2017 (ZA6861; <https://doi.org/10.4232/1.13738>).

Code availability Derived model outputs and analysis code are available as an online supplementary resource to this publication.

Authors' contributions Conceptualisation: TB, JFOM; methodology: TB, JFOM; data curation: TB; formal analysis: TB; writing—original draft: TB; writing—review and editing: BMJ, PR, JFOM; supervision: JFOM, PR. All authors contributed to the development of the manuscript and have read and approved the final version.

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